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Human Foods Program





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Toxicokinetic Chemical Simulator (ToCS): A Graphical User Interface for High- Throughput Chemical Analysis

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Office of Scientific Coordination and Computational Sciences (OSCCS)

Office of Laboratory Operations and Applied Science (OLOAS)

Session III:

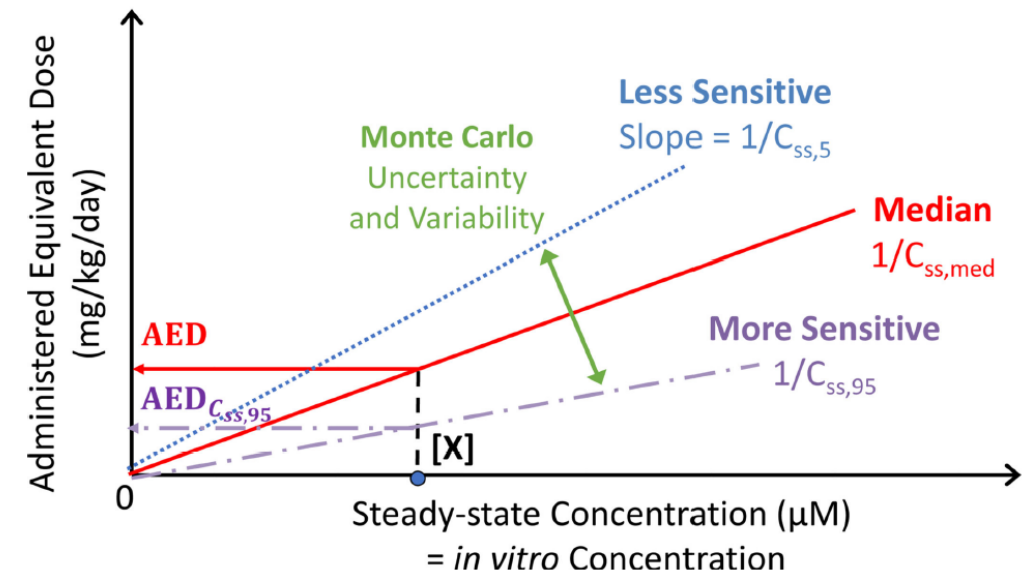
In Vitro to In Vivo Extrapolation (IVIVE)

Module

IVIVE Background

- A new approach method (NAM) that uses in vitro data to predict in vivo effects of a chemical
- Toxicokinetic IVIVE estimates the in vivo external dose (oral equivalent dose – OED) needed to produce an internal bioactive concentration comparable to that seen in vitro
- If chemical exposure data is available, then a bioactivity exposure ratio (BER) can be calculated to estimate potential health risk of the chemical
 - $BER < 1$: A chemical may pose a health risk and should be investigated further

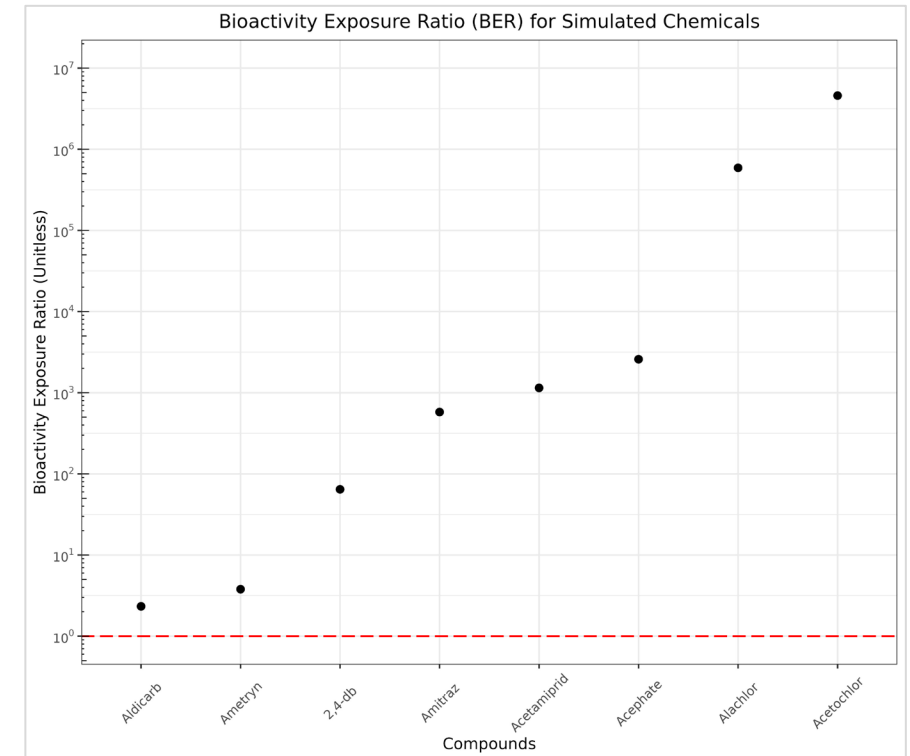
$$AED_{C_{ss,95}} = \frac{[X]}{C_{ss,95}}$$



Breen, M., Ring, C.L., Kreutz, A., Goldsmith, M.R. and Wambaugh, J.F. (2021). High-throughput PBTK models for in vitro to in vivo extrapolation. *Expert opinion on drug metabolism & toxicology*

IVIVE Module Outputs

- Plot and table of oral equivalent doses (OEDs)
 - Represents the in vivo external chemical dose needed to achieve the internal chemical concentration that causes bioactivity in vitro
- Plot and table of the bioactivity exposure ratio (BER) for each compound
 - A measure of risk, comparing current exposure level with the OED
 - Prediction is only available if exposure data is provided by the user



IVIVE Module Inputs

Required:

- Species and species preferences (if applicable)
- Model selection
- In silico parameter preference
- IVIVE conditions and output preference
- Chemicals to simulate
- In vitro bioactivity data

Optional:

- In vivo exposure data
- Bioavailability settings
- Output units, concentration type (plasma, blood), and tissue compartment
- Monte Carlo sampling parameters
- Model calculation preferences for hepatic clearance, fraction unbound, and partition coefficients

IVIVE Conditions for Selection

Assumption Selection	Metabolic Clearance		Bioactive Concentration In Vivo			Bioactive Concentration In Vitro	
	Restrictive	Non-Restrictive*	Free Venous Plasma	Total Venous Plasma	Total Specified Tissue	Nominal	Free
NULL	X				X	X	
Honda1	X		X				X
Honda2	X		X			X	
Honda3	X			X		X	
Honda4		X			X	X	

*Protein binding NOT taken into account in liver clearance

In Vitro Bioactivity Data (Required for IVIVE)

- Needed to compute an oral equivalent dose (OED)
- Data could be retrieved from:
 - EPA's CompTox Dashboard (<https://comptox.epa.gov/dashboard>)
 - NTP's ICE platform (<https://ice.ntp.niehs.nih.gov/>)
 - Internal databases
 - Literature publications

Bioactivity Data: EPA's CompTox Dashboard



B

Chemical Details

Executive Summary

Physchem Prop.

Env. Fate/Transport

Hazard Data

Safety > GHS Data

ADME > IVIVE

Exposure

Bioactivity

ToxCast: Summary

HTTr: Summary

HTPP: Summary

PubChem

ToxCast: Models

EXPORT

Bioactivity Summary Grid

Filter out non-representative sample (Repr.) results

Filter out 'background' from Intended Target Family

Name

Assay Lists

Details

SeqAPASS

Gene Symbol

AOP

Event

Repr. Plot

All Plots

Hit Call

Continuous Hit Call

Top

AC50

(1) Active

☐	CCTE_Simmons_AUR_TPO	Thyroid Bioactivity		NP_062226.2	Tpo	159 42	279			Active	1	-84.53	6.16
☐	TOX21_p450_CYP3A4_Antag	-		NP_059488.2	CYP3A4	-	-			Active	0.9983	-41.53	43.65
☐	TOX21_p450_CYP2C9_Antag	-		NP_000762.2	CYP2C9	-	-			Active	0.9551	-37.58	14.55
☐	TOX21_p450_CYP1A2_Antag	-		NP_000752.2	CYP1A2	-	-			Active	1	-58.93	20.23
☐	NVS_ENZ_oCOX2	-		NP_001009432.1	PTGS2	-	-			Active	0.9996	-33.52	15.74
☐	NVS_NR_cAR	ToxCast AR Pathway Model		NP_001009012.1	AR	23	25			Active	0.9999	-37.07	20.98
☐	NVS_NR_hPPARα	-		NP_056953.2	PPARG	-	-			Active	0.943	-44.76	25.00
☐	ATG_PXR_TRANS	-		NP_071285.1	NR1I2	-	-			Active	0.9841	1.81	38.50
☐	ATG_ERα_TRANS	ToxCast ER Pathway Model		NP_000116.2	ESR1	-	-			Active	1	3.32	41.44
☐	BSK_SAg_Eselectin	-		NP_000441.2	SELE	-	-			Active	0.9504	0.14	4.26
☐	CCTE_GLTED_hTTR_0.125uM	Thyroid Bioactivity			TTR	-	-			Active	1	102.92	26.13
☐	ATG_RXRb_TRANS	-		NP_068811.1	RXRβ	-	-			Active	0.9957	1.41	63.23
☐	CCTE_Shafer_MEA_acute_syn	-				-	-			Active	0.9686	20.62	20.00

Rows: 21 of 1,133

Total Rows: 1,133

Filtered: 21

Denominator is the total number of sample assay endpoint pairs (including non-representative samples).

Enter chemical
identifiers for the
chemical(s) you want
to find bioactivity for

Bioactivity Data: NTP's ICE Platform

Input

Results

Help

Help Video

Report an Issue

Close ICE

Curve Surfer results

Send filtered results to:

Select tool...

Select filter to add to chain:

100%

Clear Filters

Select All Filtered

Clear Selected

Select Page

1 of 38

Showing 1-10 of 376 curves.

Sort Results By

Chemical Name

Direction

Asc

Overlay Selected

Overlay Filtered

Selected Item(s): 0/376

Select this item

ACEA_AR_agonist_80hr
81-07-2

Assay: ACEA_AR_agonist_80hr
CASRN: 81-07-2
Chemical Name: Saccharin
ACS0 (uM): NA
BMD (uM): NA
Top of Curve: NA
Technological Interference: No interference flag

Mechanistic Target: Not Annotated
DTXSID: DTXSID5021251
Winning Curve-Fit Model: poly1
ACC (uM): NA
BMR: NA
Call: Flag-Omit
Flag Information: The direction of the curve does not match the intended direction of analysis

Select this item

ACEA_AR_agonist_AUC_viability
81-07-2

Assay: ACEA_AR_agonist_AUC_viability
CASRN: 81-07-2
Chemical Name: Saccharin
ACS0 (uM): NA
BMD (uM): NA
Top of Curve: NA
Technological Interference: No interference flag

Mechanistic Target: Cell Viability Process
DTXSID: DTXSID5021251
Winning Curve-Fit Model: poly1
ACC (uM): NA
BMR: NA
Call: Inactive
Flag Information: No flag information

Bioactivity Data: NTP's ICE Platform

AutoSave Off Curve_Data20250923165249-SNZSVK01U - Protected... Saved to this PC

File Home Insert Draw Page Layout Formulas Data Review View Automate Help Acrobat

Search

Comments Share

A1 m4id

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
	m4id	Assay	Direction	CASRN	DTXSID	ChemicalN	Mechanist	ModeOfc	Call	FlagInfor	AC50	ACC	BMD	Concentra	BMR	TopOfCurv	AssayResp	WinningCu	InvitroAss	Technolog	Concentra	Reference	URL
2	8362706	ACEA_AR_down	81-07-2	DTXSID502	Saccharin	Not Annot			Flag-Omit	The direct	NA	NA	NA	uM	NA	NA	percent_ar	Poly 1	Cell-based		(1:0.1:6.1E	invitroDBw	https://clowd
3	8357437	ACEA_AR_down	81-07-2	DTXSID502	Saccharin	Cell Viabili	AcuteTo		Inactive		NA	NA	NA	uM	NA	NA	percent_ar	Poly 1	Cell-based		(1:0.1:3.34	invitroDBw	https://clowd
4	8364559	ACEA_AR_up	81-07-2	DTXSID502	Saccharin	Not Annot			Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Exp 4	Cell-based		(1:0.1:-0.0	invitroDBw	https://clowd
5	8366417	ACEA_AR_down	81-07-2	DTXSID502	Saccharin	Cell Viabili	AcuteTo		Inactive		NA	NA	NA	uM	NA	NA	percent_ar	Poly 1	Cell-based		(1:0.1:-2.3	invitroDBw	https://clowd
6	8355438	ACEA_ER_down	81-07-2	DTXSID502	Saccharin	Not Annot			Flag-Omit	The direct	NA	NA	NA	uM	NA	NA	percent_ar	Gain-Loss	Cell-based		(1:0.006:-	invitroDBw	https://clowd
7	8360707	ACEA_ER_down	81-07-2	DTXSID502	Saccharin	Cell Viabili	AcuteTo		Inactive		NA	NA	NA	uM	NA	NA	percent_ar	Gain-Loss	Cell-based		(1:0.006:3	invitroDBw	https://clowd
8	12492251	ATG_AP_2_up	81-07-2	DTXSID502	Saccharin	Cell Differ	Cancer		Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Exp 2	Cell-based		(1:0.09:-0	invitroDBw	https://clowd
9	12528643	ATG_E2F_up	81-07-2	DTXSID502	Saccharin	Regulation	Cancer		Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 1	Cell-based		(1:0.09:0.1	invitroDBw	https://clowd
10	12733354	ATG_ERRa_up	81-07-2	DTXSID502	Saccharin	Steroid Ho	Cancer		Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 1	Cell-based		(1:0.09:0.0	invitroDBw	https://clowd
11	12737904	ATG_ERRg_up	81-07-2	DTXSID502	Saccharin	Steroid Ho	Cancer		Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 1	Cell-based		(1:0.09:-0	invitroDBw	https://clowd
12	12551388	ATG_FoxO_up	81-07-2	DTXSID502	Saccharin	Cell Differ	Cancer		Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 1	Cell-based		(1:0.09:-0	invitroDBw	https://clowd
13	12555937	ATG_GATA_up	81-07-2	DTXSID502	Saccharin	Regulation			Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 1	Cell-based		(1:0.09:0.1	invitroDBw	https://clowd
14	12565035	ATG_GRE_up	81-07-2	DTXSID502	Saccharin	Glucocorti	Cancer		Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 1	Cell-based		(1:0.09:-0	invitroDBw	https://clowd
15	12569584	ATG_HIF1a_up	81-07-2	DTXSID502	Saccharin	Cellular Re	AcuteTo		Flag-Omit	The poly	NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 2	Cell-based		(1:0.09:0.0	invitroDBw	https://clowd
16	12756067	ATG_HNF4_up	81-07-2	DTXSID502	Saccharin	Regulation			Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 1	Cell-based		(1:0.09:-0	invitroDBw	https://clowd
17	12574133	ATG_HNF6_up	81-07-2	DTXSID502	Saccharin	Regulation			Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 2	Cell-based		(1:0.09:-0	invitroDBw	https://clowd
18	12760944	ATG_LXRa_up	81-07-2	DTXSID502	Saccharin	Liver X Rec			Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Exp 5	Cell-based		(1:0.09:-0	invitroDBw	https://clowd
19	12765494	ATG_LXRb_up	81-07-2	DTXSID502	Saccharin	Liver X Rec			Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 1	Cell-based		(1:0.09:-0	invitroDBw	https://clowd
20	12615074	ATG_Myb_up	81-07-2	DTXSID502	Saccharin	Regulation	Cancer		Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 1	Cell-based		(1:0.09:0.0	invitroDBw	https://clowd
21	12619622	ATG_Myc_up	81-07-2	DTXSID502	Saccharin	Cell Popul	Cancer		Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 2	Cell-based		(1:0.09:-0	invitroDBw	https://clowd
22	12633269	ATG_NRF1_up	81-07-2	DTXSID502	Saccharin	Regulation			Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 2	Cell-based		(1:0.09:-0	invitroDBw	https://clowd
23	12637818	ATG_NRF2_up	81-07-2	DTXSID502	Saccharin	Cellular Re	AcuteTo		Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 2	Cell-based		(1:0.09:-0	invitroDBw	https://clowd
24	12789432	ATG_NURF_up	81-07-2	DTXSID502	Saccharin	Regulation			Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 1	Cell-based		(1:0.09:0.0	invitroDBw	https://clowd
25	12793982	ATG_PPAR_up	81-07-2	DTXSID502	Saccharin	Peroxisom	Cancer		Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 1	Cell-based		(1:0.09:-0	invitroDBw	https://clowd
26	12798531	ATG_PPAR_up	81-07-2	DTXSID502	Saccharin	Peroxisom	Cancer		Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 1	Cell-based		(1:0.09:-0	invitroDBw	https://clowd
27	12803081	ATG_PPAR_up	81-07-2	DTXSID502	Saccharin	Peroxisom	Cancer		Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Exp 4	Cell-based		(1:0.09:0.2	invitroDBw	https://clowd
28	12651465	ATG_Pax6_up	81-07-2	DTXSID502	Saccharin	Cell Devel	Cancer		Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 1	Cell-based		(1:0.09:-0	invitroDBw	https://clowd
29	12830381	ATG_RORg_up	81-07-2	DTXSID502	Saccharin	Regulation	Cancer		Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 1	Cell-based		(1:0.09:-0	invitroDBw	https://clowd
30	12683308	ATG_SREB_up	81-07-2	DTXSID502	Saccharin	Cellular Re	AcuteTo		Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 1	Cell-based		(1:0.09:0)	invitroDBw	https://clowd
31	12687857	ATG_STAT_up	81-07-2	DTXSID502	Saccharin	Regulation	Cancer		Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 2	Cell-based		(1:0.09:-0	invitroDBw	https://clowd
32	12674210	ATG_Sav_up	81-07-2	DTXSID502	Saccharin	Regulation			Inactive		NA	NA	NA	uM	NA	NA	log2_fold_j	Poly 2	Cell-based		(1:0.09:0	invitroDBw	https://clowd

Export MetaData Query

Ready Display Settings 100%

Determine the bioactive concentration for each compound by exploring only “active” calls and using your desired measure and calculation method (generally use the AC50 or ACC)

Uploading a CSV File of Bioactivity Data

Toxicokinetic Chemical Simulator (ToCS)
General Parameters
Model Specifications
Compound Selection
Advanced (Optional) Parameters
Run Simulation

INSTRUCTIONS

You must choose at least one compound from the preloaded compounds, upload a CSV file with data for at least one compound not included in the preloaded compounds, or both.

Click on the appropriate link(s) below to download guidance on how to upload data under the 'Uploaded Data' card. Follow the 'Instructions' document in the downloaded folder to correctly format the file you want to upload.

[Uploaded Physical-Chemical Data File Folder](#)

[Bioactivity Data File Folder](#)

[Exposure Data File Folder](#)

PRELOADED COMPOUNDS

Select an IVIVE assumption to implement. For any input nominal bioactive concentration in vitro, the Honda1 assumption is recommended. Leave on 'NULL' if no assumptions are to be applied. See the 'IVIVE Simulation Examples' vignette for the description of the below assumption categories.

NULL

Select the types of compounds you want to simulate.

Choose from all available chemicals

Select any preloaded compounds. Search through the list by clicking on the box and scrolling or typing in a name. The list may not show all available compounds. Click on a compound to select it. You may select multiple.

At least one compound must be selected or uploaded

UPLOADED DATA

Upload a CSV file of physical and chemical data for compounds not in the preloaded list (if desired). Download the 'Uploaded Physical-Chemical Data File Folder' under the 'Instructions' card for file formatting instructions.

Browse... No file selected

Upload a CSV file with in vitro bioactive concentrations (uM units) for all selected compounds. Download the 'Bioactivity Data File Folder' under the 'Instructions' card for file formatting instructions.

Browse... No file selected

Required

Upload a CSV file of exposure data for all selected compounds (optional). Download the 'Exposure Data File Folder' under the 'Instructions' card for file formatting instructions.

Browse... No file selected

Downloadable folder with instructions for uploading bioactivity data CSV

Click 'Browse' to find and upload the CSV file of bioactivity data

Uploading a CSV File of Bioactivity Data

Toxicokinetic Chemical Simulator (ToCS) General Parameters Model Specifications

INSTRUCTIONS

You must choose at least one compound from the preloaded compounds, upload a CSV file with data for at least one compound not included in the preloaded compounds, or both.

Click on the appropriate link(s) below to download guidance on how to upload data under the 'Uploaded Data' card. Follow the 'Instructions' document in the downloaded folder to correctly format the file you want to upload.

[Uploaded Physical-Chemical Data File Folder](#)

[Bioactivity Data File Folder](#)

[Exposure Data File Folder](#)

Instructions
Type: Microsoft Word Document

SampleBioactivityData
Type: Microsoft Excel Comma Separated Values File

Instructions for Uploading Bioactivity Data

1. Take note of the chemical names and CAS numbers of the chemicals you selected to simulate. These same identifiers must be used in the bioactivity data file uploaded.
2. Prepare your CSV file to upload with bioactivity data. The file must be in the exact same format as the 'SampleBioactivityData.csv' file available in the downloaded folder. It is recommended that you copy the 'SampleBioactivityData.csv' file into a new file and edit the information to suit your simulation. The file must have the same column names and be in the same order as the sample file. There should be no empty cells in the file that you upload – every chemical simulated must have their chemical name, CAS number, and bioactive concentration (uM units, typically AC50 or ACC values) identified.
3. Once you have created your CSV file, click the 'Browse' button on the interface and select your CSV file.

	A	B	C	D	E
1	ChemicalName	CAS	BioactiveConcentration		
2	Chem1	111-11-1	1.2		
3	Chem2	222-22-2	50.3		
4	Chem3	333-33-3	22.9		
5	Chem4	444-44-4	0.95		

Uploading a CSV File of Bioactivity Data

- Copy and paste the SampleBioactivityData.csv file into the location of choice and rename as desired.
- Leave the column headers as is. Delete the entries in all other rows.
- Fill in each chemical name that you will be simulating and their corresponding CAS numbers. These identifiers **MUST** match the names and CAS numbers from the preloaded list or uploaded chemical data.
- Fill in the in vitro bioactive concentration (in uM units) for each compound. Only one bioactive concentration per compound is permitted.
- Save the CSV file.
- Upload CSV file to ToCS interface.

E17						
	A	B	C	D	E	
1	ChemicalName	CAS	BioactiveConcentration			
2	Chem1	111-11-1	1.2			
3	Chem2	222-22-2	50.3			
4	Chem3	333-33-3	22.9			
5	Chem4	444-44-4	0.95			

In Vivo Exposure Data (Optional)

- Needed to compute the bioactivity exposure ratio (BER)
- Data could be retrieved from:
 - EPA's CompTox Dashboard (<https://comptox.epa.gov/dashboard>)
 - NTP's ICE platform (<https://ice.ntp.niehs.nih.gov/>)
 - FDA's Cumulative Estimated Daily Intake (CEDI) public database (<https://www.hfpappexternal.fda.gov/scripts/fdcc/index.cfm?set=CEDI>)

Exposure Data: EPA's CompTox Dashboard

Chemical Details

Executive Summary

Physchem Prop.

Env. Fate/Transport

Hazard Data

Safety > GHS Data

ADME > IVIVE

Exposure

Product & Use Categories

Chemical Weight Fraction

Chemical Functional Use

Toxics Release Inventory

Biomonitoring Data

Exposure Predictions

Production Volume

National Health and Nutrition Examination Survey (NHANES) Inferences (mg/kg-bw/day)

Search Monitoring Data

Monitoring Data

Demographic	Median	Upper Bound (Median)	Lower Bound (Median)
Age 6-11	4.33e-5	4.92e-5	3.80e-5
Age 12-19	2.93e-5	3.38e-5	2.55e-5
Age 20-65	3.02e-5	3.27e-5	2.79e-5
Age 65+	2.10e-5	2.31e-5	1.91e-5
BMI <= 30	3.16e-5	3.30e-5	3.02e-5
BMI > 30	2.55e-5	2.74e-5	2.38e-5
Females	2.80e-5	3.03e-5	2.58e-5
Males	3.15e-5	3.37e-5	2.94e-5
Repro. Age Females	3.06e-5	3.31e-5	2.83e-5
Total	2.97e-5	3.08e-5	2.86e-5

National Health and Nutrition Examination Survey (NHANES) Data

In Silico Predicted Data (EPA Models)

Chemical Details

Executive Summary

Physchem Prop.

Env. Fate/Transport

Hazard Data

Safety > GHS Data

ADME > IVIVE

Exposure

Product & Use Categories

Chemical Weight Fraction

Chemical Functional Use

Toxics Release Inventory

Biomonitoring Data

Exposure Predictions

Production Volume

Exposure - Exposure Predictions (mg/kg-bw/day)

Search Demographics Predictions Data

Demographics Predictions Data

Demographic	Predictor	Median	Upper 95%ile	Units
	(2) SEEM2 Heuristic, SEEM3 Con			
Age 6-11	SEEM2 Heuristic	6.30e-5	1.05e-2	mg/kg/day
Age 12-19	SEEM2 Heuristic	5.87e-5	1.72e-2	mg/kg/day
Age 20-65	SEEM2 Heuristic	5.68e-5	1.15e-2	mg/kg/day
Age 66+	SEEM2 Heuristic	6.61e-5	1.95e-2	mg/kg/day
BMI <= 30	SEEM2 Heuristic	6.25e-5	1.36e-2	mg/kg/day
BMI > 30	SEEM2 Heuristic	7.07e-5	1.86e-2	mg/kg/day
Females	SEEM2 Heuristic	1.24e-5	2.90e-3	mg/kg/day
Males	SEEM2 Heuristic	3.87e-5	6.31e-3	mg/kg/day

Exposure Data: NTP's ICE Platform

Search >

Chemical Quest >

Curve Surfer >

PBPK >

IVIVE >

Chemical Characterization >

Data >

Help Videos >

Integrated Chemical Environment

Input

Results

The Search tool allows you to query ICE data using chemical quick list selections (chemical selection of data sets organized by toxicity endpoints of regulatory interest).

Run Reset

Scope of Search: Union

Chemical Input

Select Chemicals 1 chemical quick list selected.

Quick List CASRNs

82657-04-3
103-90-2
50-28-2
56-23-5
80-05-7

User Chemical Identifiers

Select Data Sets

Select Data Sets

Data Set

☒ SEEM3, Exposure Predictions

Integrated Chemical Environment

Input

Results

Search Results

> Selected Chemical Quick Lists (1)

> Selected Assays (1)

> Chemical Identifiers Not Returned By Query (5)

404-86-4, 15663-27-1, 50-28-2, 7782-49-2, 59-05-2

Chemical Identifiers Not Returned By Query

View Data Tables

Data Table Data Summary Download

Interact with Results

To zoom in to a specific data type, click on the box within the map, click on the banner. As you navigate through the map, position in the map.

Details

Clear Filter

Endpoint Record Count: 55

Record ID	Chemical Name	Substance Type	CASRN	DTXSID	QSAR Ready ID	Assay	Endpoint	Response	Unit
exposure_...	Acetaminophen	Chemical	103-90-2	DTXSID2020006	RZVAJINK... UHFFFAO... N	SEEM3, Exposure Predictions	95th percentile	1.889E-4	mg/kg/day
exposure_...	Acetaminophen	Chemical	103-90-2	DTXSID2020006	RZVAJINK... UHFFFAO... N	SEEM3, Exposure Predictions	5th percentile	3.677E-9	mg/kg/day

Exposure Data: FDA CEDI Database

Be careful of units! ToCS needs mg/kg-BW/day.

**U.S. FOOD & DRUG
ADMINISTRATION**

Cumulative Estimated Daily Intake (CEDI)

[FDA Home](#)
[Food Ingredient and Packaging Inventories](#)
[CEDI Database](#)
[Cumulative Estimated Daily Intake \(CEDI\)](#)

Database updated May 2024

As part of the premarket notification process for food contact substances (FCSs), the Office of Food Additive Safety (OFAS) maintains a database of cumulative estimated daily intakes (CEDIs) for a large number of FCSs. This database is referred to as the CEDI database and is publicly available. FDA updates the CEDI database as additional information becomes available.

The CEDIs in the database were calculated using information currently available at the time and may be subject to revision on the basis of new information as it is made available and evaluated or existing information is re-evaluated by OFAS.

Information for FCS Notifiers

As the safety evaluation is the responsibility of the notifier, potential notifiers are encouraged to approach OFAS through a prenotification consultation to determine if their proposed use is supported by the available data. This is especially critical if the current CEDI may change substantially as a result of the intended use. Notifiers are also encouraged to contact OFAS with new information that may be used to calculate CEDIs and include such information in notifications. See [Preparation of Food Contact Notifications and Food Additive Petitions for Food Contact Substances: Chemistry Recommendations](#). Concerning the database, OFAS notes the following:

- The CEDI values listed in the database are primarily for FCSs that are regulated under 21 CFR 175-178 and/or authorized by a Food Contact Substance Notification. Information on constituents, such as monomers, and other impurities may not currently be available.
- The listed CEDI values for polymeric FCSs in the database are predominately for exposure to the associated low molecular weight oligomers.
- The CEDI values are expressed as both cumulative dietary concentration (CDC, parts-per-billion, ppb) and as cumulative estimated daily intake (CEDI, microgram/kilogram body weight/person/day, µg/kg bw/d).

Search and display hints:

- Select the specific substance below to view additional information (other names) for that substance.
- Select the specific regulation (21 CFR), to view the text of that regulation.
- To sort by a specific field, click on the column header for that field.
- To browse the records, use the Show All, First/Previous/Next/Last, and Jump To options at the bottom of the data table.
- To search for a specific substance/term, enter the term in the Search box and select Show Items to display only those records that contain the selected term. (The search results also includes terms not shown on this page, but included in the full record on the detail page.)
- The search results will return hits of records containing words that include the search term. For example, a search for the color **red** will return results that include terms such as **reduce**, **ingredient**, and **cultured**. To limit results to only the searched term, place a space before and after the word in the basic search or in the advanced search "this exact phrase" field.

Download data from this searchable database in Excel format. If you need help accessing information in different file formats, see [Instructions for Downloading Viewers and Players](#).

Basic Search Advanced Search Field Search						
Search:			Show Items Clear		Records Found: 1252 Show All Page 1 of 26	
CAS Reg. No.* (or other ID*)	Substance* (sorted A-Z)	CEDI* (µg/kg bw/d)	CDC* (ppb)	Calculation/update* Date	21 CFR*	
78-63-7	(1,1,4,4-TETRAMETHYLTETRAMETHYLENE)BIS(TERT-BUTYL PEROXIDE)	0.01	0.2	Jun 17, 2010	177.1520 177.2600	
68259-36-9	(2,4,4-TRIMETHYLPENT-2-YL)-N-PHENYL-1-NAPHTHYLAMINE	2.5	50	Jul 24, 2007		
6891-44-7	(2-(METHACRYLOYLOXY)ETHYL)TRIMETHYLAMMONIUM METHYL SULFATE	0.02	0.4	Sep 6, 2000	173.10 178.3520	
68784-12-3	(2-ALKENYL(C15-21))SUCCINIC ANHYDRIDE	2.6	52	Oct 23, 2015	176.170	
26401-86-5	(N-OCTYL)TIN S,S',S"-TRIS(ISOCTYLMERCAPTOACETATE)	0.385	7.7	Jul 22, 1985	178.2650	
4118-16-5	1,1'-(6-PHENYL-1,3,5-TRIAZINE-2,4-DIYL)DIIMINO)BIS[9,10-ANTHRACENEDIONE]	0.025	0.5	Mar 21, 1985	178.3297	
94-03-1	1,1'-OXYBIS(2-PROPANOL) DIBENZOATE	0.85	17	Jun 20, 2000		
1163733-25-2	1,1,2,2-TETRAFLUORO-2-[(1,2,2-TRIFLUOROETHENYL)OXY]ETHANESULFONYL FLUORIDE, POLYMER WITH 1,1,2,2-TETRAFLUOROETHENE, HYDROLYZED	0.0000265	0.00053	Nov 13, 2017		
1843-03-4	1,1,3-TRIS(5-TERT-BUTYL-4-HYDROXY-2-METHYLPHENYL)BUTANE	47	940	Nov 7, 1983	175.105 178.2010	
75-37-6	1,1-DIFLUOROETHANE	0.55	11	Oct 15, 1992	178.3010 701.30	
75-38-7	1,1-DIFLUOROETHENE	0.000035	0.0007	Feb 29, 2000	177.1350 177.1380 177.2510 177.2600	
64253-30-1	1,11-(3,6,9-TRIOXAUNDECYL) BIS[3-(DODECYLTHIO)PROPIONATE]	0.6	12	Sep 26, 1996	178.2010	
887192-39-4	1,12,23,34-TETRAAZACYCLOTETRADECANONE-2,11,24,33-TETRAONE	0.11	2.2	Aug 19, 2014		

Uploading a CSV File of Exposure Data

Toxicokinetic Chemical Simulator (ToCS)

General Parameters Model Specifications **Compound Selection** Advanced (Optional) Parameters Run Simulation

INSTRUCTIONS

You must choose at least one compound from the preloaded compounds, upload a CSV file with data for at least one compound not included in the preloaded compounds, or both.

Click on the appropriate link(s) below to download guidance on how to upload data under the 'Uploaded Data' card. Follow the 'Instructions' document in the downloaded folder to correctly format the file you want to upload.

[Uploaded Physical-Chemical Data File Folder](#)

[Bioactivity Data File Folder](#)

[Exposure Data File Folder](#)

PRELOADED COMPOUNDS

Select an IVIVE assumption to implement. For any input nominal bioactive concentration in vitro, the Honda1 assumption is recommended. Leave on 'NULL' if no assumptions are to be applied. See the 'IVIVE Simulation Examples' vignette for the description of the below assumption categories.

NULL

Select the types of compounds you want to simulate.

Choose from all available chemicals

Select any preloaded compounds. Search through the list by clicking on the box and scrolling or typing in a name. The list may not show all available compounds. Click on a compound to select it. You may select multiple.

At least one compound must be selected or uploaded

UPLOADED DATA

Upload a CSV file of physical and chemical data for compounds not in the preloaded list (if desired). Download the 'Uploaded Physical-Chemical Data File Folder' under the 'Instructions' card for file formatting instructions.

Browse... No file selected

Upload a CSV file with in vitro bioactive concentrations (uM units) for all selected compounds. Download the 'Bioactivity Data File Folder' under the 'Instructions' card for file formatting instructions.

Browse... No file selected

Required

Upload a CSV file of exposure data for all selected compounds (optional). Download the 'Exposure Data File Folder' under the 'Instructions' card for file formatting instructions.

Browse... No file selected

Downloadable folder with instructions for uploading exposure data CSV

Click 'Browse' to find and upload the CSV file of exposure data

Uploading a CSV File of Exposure Data

Toxicokinetic Chemical Simulator (ToCS)
General Parameters
Model Spe

INSTRUCTIONS

You must choose at least one compound from the preloaded compounds, upload a CSV file with data for at least one compound not included in the preloaded compounds, or both.

Click on the appropriate link(s) below to download guidance on how to upload data under the 'Uploaded Data' card. Follow the 'Instructions' document in the downloaded folder to correctly format the file you want to upload.

[Uploaded Physical-Chemical Data File Folder](#)

[Bioactivity Data File Folder](#)

[Exposure Data File Folder](#)

Instructions for Uploading Exposure Data

1. Take note of the chemical names and CAS numbers of the chemicals you selected to simulate. These same identifiers must be used in the exposure data file uploaded.
2. Prepare your CSV file to upload with exposure data. The file must be in the exact same format as the 'SampleExposureData.csv' file available in the downloaded folder. It is recommended that you copy the 'SampleExposureData.csv' file into a new file and edit the information to suit your simulation. The file must have the same column names and be in the same order as the sample file. There should be no empty cells in the ChemicalName or CAS columns, and at least one exposure estimate (mg/kg BW/day) per chemical must be uploaded (either an upper, median, or lower exposure estimate).
3. Once you have created your CSV file, click the 'Browse' button on the interface and select your CSV file.

Instructions
Type: Microsoft Word Document

SampleExposureData
Type: Microsoft Excel Comma Separated Values File

	A	B	C	D	E	F	G
1	ChemicalName	CAS	Upper	Median	Lower		
2	Chem1	111-11-1	2.39E-04	5.99E-07			
3	Chem2	222-22-2	1.29E-04	1.31E-06			
4	Chem3	333-33-3	1.44E-06	3.37E-07	4.79E-08		
5	Chem4	444-44-4	2.11E-07	4.93E-08	4.74E-09		
6							

Uploading a CSV File of Exposure Data

- Copy and paste the SampleExposureData.csv file into the location of choice and rename as desired.
- Leave the column headers as is. Delete the entries in all other rows.
- Fill in each chemical name that you will be simulating and their corresponding CAS numbers. These identifiers **MUST** match the names and CAS numbers from the preloaded list or uploaded chemical data.
- Fill in the upper, median, and lower exposure estimates (mg/kg-BW/day) as desired. There must be **AT LEAST** one exposure point per compound.
- Save the CSV file.
- Upload CSV file to ToCS interface.

I19							
	A	B	C	D	E	F	G
1	ChemicalName	CAS	Upper	Median	Lower		
2	Chem1	111-11-1	2.39E-04	5.99E-07			
3	Chem2	222-22-2	1.29E-04	1.31E-06			
4	Chem3	333-33-3	1.44E-06	3.37E-07	4.79E-08		
5	Chem4	444-44-4	2.11E-07	4.93E-08	4.74E-09		
6							

Example 1 Scenario

- Obtain the bioactivity exposure ratios of six flavoring agents using the pbtk model

Chemical Name	CAS	Bioactive Concentration*	Exposure Upper^	Exposure Median^	Exposure Lower^
Butylparaben	94-26-8	14.06	1.51E-6	1.28E-6	1.09E-6
Acetophenone	98-86-2	15.90	0.263	6.07E-5	
Methyl salicylate	119-36-8	31.40	0.184	2.17E-4	
Resorcinol	108-46-3	6.626	5.85E-2	1.91E-5	
o-Cresol	95-48-7	47.30	0.105	3.22E-4	
p-Cresol	106-44-5	38.88	1.52	1.53E-3	

- Do not use the available in silico parameters
- Return the 5th OED quantile (95th steady state concentration quantile)
- Use the Honda1 IVIVE assumption

* = uM
^ = mg/kw-BW/day

Example 1: General Parameters Tab

Toxicokinetic Chemical Simulator (ToCS)
General Parameters
Model Specifications
Compound Selection
Advanced (Optional) Parameters
Run Simulation

INSTRUCTIONS

Fill out the prompts on each of the above tabs moving left to right. Then, click the 'Run Simulation' tab to run the simulation or reset all selections.

ToCS provides four outputs: 1) Concentration-time profiles (returns chemical concentrations in body compartments over time), 2) Steady state (SS) concentration (returns SS concentrations in body compartments from an oral infusion), 3) In vitro to in vivo extrapolation (IVIVE) (returns oral equivalent doses to in vitro bioactive concentrations), 4) Parameter calculations (returns elimination rates, volumes of distribution, tissue to unbound plasma partition coefficients, half-lives, and total plasma clearances).

This application uses the U.S. EPA's R package 'httk'. For more information on ToCS and 'httk', please refer to the following links.

[Vignettes \(ToCS tutorials\)](#)

[Report ToCS issues/suggestions](#)

[httk publication](#)

[httk CRAN webpage](#)

OUTPUT

Select the desired output.

In vitro in vivo extrapolation (IVIVE)

SPECIES

Select the species to analyze.

Human

Example 1: Model Specifications Tab

Toxicokinetic Chemical Simulator (ToCS)
General Parameters
Model Specifications
Compound Selection
Advanced (Optional) Parameters
Run Simulation

MODEL

Select the model to simulate. If a species other than 'Human' is selected, '3compartmentss' must be chosen.

pbtk

Select whether to use in silico generated parameters for compounds with missing in vitro data. These parameters will not overwrite existing in vitro data, and it will expand the number of compounds available.

No, do not load in silico parameters

Select whether to return all oral equivalent dose (OED) samples for each compound or a selected quantile.

Only return a specified dose quantile (default)

Enter the steady state concentration quantile (as a decimal) to be used in the OED calculation. Selecting the 95th concentration quantile will output the 5th OED quantile.

0.95

DOSING

No options for this category.

Example 1: Compound Selection Tab

Toxicokinetic Chemical Simulator (ToCS)
General Parameters
Model Specifications
Compound Selection
Advanced (Optional) Parameters
Run Simulation

INSTRUCTIONS

Click on the appropriate link(s) below to download guidance on how to upload data under the 'Uploaded Data' card. Follow the 'Instructions' document in the downloaded folder to correctly format the file you want to upload.

[Uploaded Physical-Chemical Data File Folder](#)

[Bioactivity Data File Folder](#)

[Exposure Data File Folder](#)

PRELOADED COMPOUNDS

Select an IVIVE assumption to implement. For any input nominal bioactive concentration in vitro, the Honda1 assumption is recommended. Leave on 'NULL' if no assumptions are to be applied. See the 'IVIVE Simulation Examples' vignette for the description of the below assumption categories.

Honda1

Select the types of compounds you want to simulate.

Choose from all available chemicals

Enter the volume fraction of fetal bovine serum used in the in vitro assay.

0.1

Select any preloaded compounds. Search through the list by clicking on the box and scrolling or typing in a name. The list may not show all available compounds. Click on a compound to select it. You may select multiple.

94-26-8, Butylparaben

98-86-2, Acetophenone

119-36-8, Methyl salicylate

108-46-3, Resorcinol

95-48-7, 2-methylphenol

106-44-5, 4-methylphenol

UPLOADED DATA

Upload a CSV file of physical and chemical data for compounds not in the preloaded list (if desired). Download the 'Uploaded Physical-Chemical Data File Folder' under the 'Instructions' card for file formatting instructions.

Browse...

No file selected

Upload a CSV file with in vitro bioactive concentrations (uM units) for all selected compounds. Download the 'Bioactivity Data File Folder' under the 'Instructions' card for file formatting instructions.

Browse...

BioactivityDataEx1.csv

Upload complete

Upload a CSV file of exposure data for all selected compounds (optional). Download the 'Exposure Data File Folder' under the 'Instructions' card for file formatting instructions.

Browse...

ExposureDataEx1.csv

Upload complete

Example 1: Bioactivity and Exposure Data Files

Excel interface for **BioactivityDataEx1**. The ribbon shows **File**, **Home**, **Insert**, **Draw**, **Page Layout**, and **Formulas**. The **Home** ribbon is active, showing the **Clipboard**, **Font**, and **Align** groups. The font is set to **Aptos Narrow** size **11**. The active cell is **D9**.

	A	B	C	D	E
1	ChemicalName	CAS	BioactiveConcentration		
2	Butylparaben	94-26-8	14.06		
3	Acetophenone	98-86-2	15.9		
4	Methyl salicylate	119-36-8	31.4		
5	Resorcinol	108-46-3	6.626		
6	2-methylphenol	95-48-7	47.3		
7	4-methylphenol	106-44-5	38.88		

Excel interface for **ExposureDataEx1**. The ribbon shows **File**, **Home**, **Insert**, **Draw**, **Page Layout**, and **Formulas**. The **Home** ribbon is active, showing the **Clipboard**, **Font**, and **Align** groups. The font is set to **Aptos Narrow** size **11**. The active cell is **D23**.

	A	B	C	D	E	F
1	ChemicalName	CAS	Upper	Median	Lower	
2	Butylparaben	94-26-8	1.51E-06	1.28E-06	1.09E-06	
3	Acetophenone	98-86-2	0.263	6.07E-05		
4	Methyl salicylate	119-36-8	0.184	2.17E-04		
5	Resorcinol	108-46-3	5.85E-02	1.91E-05		
6	2-methylphenol	95-48-7	0.105	3.22E-04		
7	4-methylphenol	106-44-5	1.52	1.53E-03		
8						

Example 1: Advanced Parameters Tab

Toxicokinetic Chemical Simulator (ToCS)
General Parameters
Model Specifications
Compound Selection
Advanced (Optional) Parameters
Run Simulation

MODEL CONDITIONS

Enter the number of Monte Carlo samples generated for each compound.

Select whether to adjust the chemical fraction unbound in presence of plasma proteins for lipid binding.

Yes, adjust the fraction of unbound plasma (default) ▼

Select whether to use regressions when calculating partition coefficients.

Use regressions (default) ▼

Enter the p-value threshold for the in vitro intrinsic hepatic clearance rate where clearance assay results with p-values above this threshold are set to zero.

Enter the minimum acceptable chemical fraction unbound in presence of plasma proteins. All values below this will be set to this value.

MODEL SOLVER

No options for this category.

BIOAVAILABILITY

Enter a default value for the Caco-2 apical-to-basal membrane permeability (denoted Caco2.Pab, 10⁻⁶ cm/s).

Select whether to use the Caco2.Pab value set above to estimate F_{abs} (the in vivo measured fraction of an oral dose absorbed from the gut lumen into the gut) if bioavailability data is unavailable.

Use the Caco2.Pab value selected above (default) ▼

Select whether to use the Caco2.Pab value set above to calculate F_{gut} (the in vivo measured fraction of an oral dose that passes gut metabolism and clearance) if bioavailability data is unavailable.

Use the Caco2.Pab value selected above (default) ▼

Select whether to overwrite in vivo F_{abs} and F_{gut} data (if available).

Do not overwrite in vivo values (default) ▼

Select whether to keep F_{abs} and F_{gut} at 100% availability (which overwrites all other bioavailability parameter settings above).

Do not keep Fabs and Fgut at 100% availability (default) ▼

OUTPUT SPECIFICATION

Select the dose output units from either mg/kg BW/day (mgpkgsday) (default) or umol/kg BW/day (umolpkgsday).

mgpkgsday ▼

Human Foods Program

Human Foods Program

Human Foods Program

Human Foods Program

Human Foods Program

Example 2 Scenario

- Obtain the oral equivalent doses (OEDs) of four direct food additives using the pbtk model

Chemical Name	CAS	Bioactive Concentration*	Exposure Upper^	Exposure Median^	Exposure Lower^
Saccharin	81-07-2	4.324E-3	1.24	2.02E-7	
Theobromine	83-67-0	5.316	1.73E-2	4.19E-6	
Benzophenone	119-61-9	4.425	0.236	3.09E-4	
Eugenol	97-53-0	16.01	0.353	9.20E-5	

* = uM
^ = mg/kw-BW/day

- Do not use the available in silico parameters
- Run 500 Monte Carlo steady state samples
- Return the all OED samples
- Use the Honda1 IVIVE assumption

Example 2: General Parameters Tab

Toxicokinetic Chemical Simulator (ToCS)
General Parameters
Model Specifications
Compound Selection
Advanced (Optional) Parameters
Run Simulation

INSTRUCTIONS

Fill out the prompts on each of the above tabs moving left to right. Then, click the 'Run Simulation' tab to run the simulation or reset all selections.

ToCS provides four outputs: 1) Concentration-time profiles (returns chemical concentrations in body compartments over time), 2) Steady state (SS) concentration (returns SS concentrations in body compartments from an oral infusion), 3) In vitro to in vivo extrapolation (IVIVE) (returns oral equivalent doses to in vitro bioactive concentrations), 4) Parameter calculations (returns elimination rates, volumes of distribution, tissue to unbound plasma partition coefficients, half-lives, and total plasma clearances).

This application uses the U.S. EPA's R package 'httk'. For more information on ToCS and 'httk', please refer to the following links.

[Vignettes \(ToCS tutorials\)](#)

[Report ToCS issues/suggestions](#)

[httk publication](#)

[httk CRAN webpage](#)

OUTPUT

Select the desired output.

In vitro in vivo extrapolation (IVIVE)

SPECIES

Select the species to analyze.

Human

Example 2: Model Specifications Tab

Toxicokinetic Chemical Simulator (ToCS)
General Parameters
Model Specifications
Compound Selection
Advanced (Optional) Parameters
Run Simulation

MODEL

Select the model to simulate. If a species other than 'Human' is selected, '3compartmentss' must be chosen.

pbt▼

Select whether to use in silico generated parameters for compounds with missing in vitro data. These parameters will not overwrite existing in vitro data, and it will expand the number of compounds available.

No, do not load in silico parameters▼

Select whether to return all oral equivalent dose (OED) samples for each compound or a selected quantile.

Return all OED samples (will also return the 5th dose quantile)▼

DOSING

No options for this category.

Example 2: Compound Selection Tab

Toxicokinetic Chemical Simulator (ToCS)
General Parameters
Model Specifications
Compound Selection
Advanced (Optional) Parameters
Run Simulation

INSTRUCTIONS

Click on the appropriate link(s) below to download guidance on how to upload data under the 'Uploaded Data' card. Follow the 'Instructions' document in the downloaded folder to correctly format the file you want to upload.

[Uploaded Physical-Chemical Data File Folder](#)

[Bioactivity Data File Folder](#)

[Exposure Data File Folder](#)

PRELOADED COMPOUNDS

Select an IVIVE assumption to implement. For any input nominal bioactive concentration in vitro, the Honda1 assumption is recommended. Leave on 'NULL' if no assumptions are to be applied. See the 'IVIVE Simulation Examples' vignette for the description of the below assumption categories.

Honda1

Select the types of compounds you want to simulate.

Choose from all available chemicals

Enter the volume fraction of fetal bovine serum used in the in vitro assay.

0.1

Select any preloaded compounds. Search through the list by clicking on the box and scrolling or typing in a name. The list may not show all available compounds. Click on a compound to select it. You may select multiple.

81-07-2, Saccharin
83-67-0, Theobromine
119-61-9, Benzophenone
97-53-0, Eugenol

UPLOADED DATA

Upload a CSV file of physical and chemical data for compounds not in the preloaded list (if desired). Download the 'Uploaded Physical-Chemical Data File Folder' under the 'Instructions' card for file formatting instructions.

Browse...
No file selected

Upload a CSV file with in vitro bioactive concentrations (uM units) for all selected compounds. Download the 'Bioactivity Data File Folder' under the 'Instructions' card for file formatting instructions.

Browse...
BioactivityDataEx2.csv
Upload complete

Upload a CSV file of exposure data for all selected compounds (optional). Download the 'Exposure Data File Folder' under the 'Instructions' card for file formatting instructions.

Browse...
ExposureDataEx2.csv
Upload complete

Example 2: Bioactivity and Exposure Data

AutoSave Off BioactivityDataEx2

File Home Insert Draw Page Layout Formulas Data

Paste Clipboard Aptos Narrow 11 B I U Font

C9

	A	B	C	D	E
1	ChemicalName	CAS	BioactiveConcentration		
2	Saccharin	81-07-2	4.32E-03		
3	Theobromine	83-67-0	5.316		
4	Benzophenone	119-61-9	4.425		
5	Eugenol	97-53-0	16.01		
6					

AutoSave Off ExposureDataEx2

File Home Insert Draw Page Layout Formulas Data

Paste Clipboard Aptos Narrow 11 B I U Font

J14

	A	B	C	D	E	F	G
1	ChemicalName	CAS	Upper	Median	Lower		
2	Saccharin	81-07-2	1.24	2.02E-07			
3	Theobromine	83-67-0	1.73E-02	4.19E-06			
4	Benzophenone	119-61-9	0.236	3.09E-04			
5	Eugenol	97-53-0	0.353	9.20E-05			
6							

Example 2: Advanced Parameters Tab

Toxicokinetic Chemical Simulator (ToCS)
General Parameters
Model Specifications
Compound Selection
Advanced (Optional) Parameters
Run Simulation

MODEL CONDITIONS

Enter the number of Monte Carlo samples generated for each compound.

Select whether to adjust the chemical fraction unbound in presence of plasma proteins for lipid binding.

Yes, adjust the fraction of unbound plasma (default) ▼

Select whether to use regressions when calculating partition coefficients.

Use regressions (default) ▼

Enter the p-value threshold for the in vitro intrinsic hepatic clearance rate where clearance assay results with p-values above this threshold are set to zero.

Enter the minimum acceptable chemical fraction unbound in presence of plasma proteins. All values below this will be set to this value.

MODEL SOLVER

No options for this category.

BIOAVAILABILITY

Enter a default value for the Caco-2 apical-to-basal membrane permeability (denoted Caco2.Pab, 10⁻⁶ cm/s).

Select whether to use the Caco2.Pab value set above to estimate F_{abs} (the in vivo measured fraction of an oral dose absorbed from the gut lumen into the gut) if bioavailability data is unavailable.

Use the Caco2.Pab value selected above (default) ▼

Select whether to use the Caco2.Pab value set above to calculate F_{gut} (the in vivo measured fraction of an oral dose that passes gut metabolism and clearance) if bioavailability data is unavailable.

Use the Caco2.Pab value selected above (default) ▼

Select whether to overwrite in vivo F_{abs} and F_{gut} data (if available).

Do not overwrite in vivo values (default) ▼

Select whether to keep F_{abs} and F_{gut} at 100% availability (which overwrites all other bioavailability parameter settings above).

Do not keep Fabs and Fgut at 100% availability (default) ▼

OUTPUT SPECIFICATION

Select the dose output units from either mg/kg BW/day (mgpkpday) (default) or umol/kg BW/day (umolpkpday).

mgpkpday ▼

Human Foods Program



Human Foods Program

Human Foods Program

Example 2: Run Simulation Tab

Toxicokinetic Chemical Simulator (ToCS)

General Parameters Model Specifications Compound Selection Advanced (Optional) Parameters **Run Simulation**

ACTIONS

Click on the 'Run Simulation' button when all information has been entered.

Run Simulation

Click on the button below to reset your session. This will clear all selections and any uploaded data, and is recommended to be done every time a new simulation is run.

Reset Session

☒ Check the box to display plots with a log10 scale y-axis.

SELECTED COMPOUNDS

Selected_Compounds

Benzophenone

Eugenol

Saccharin

Theobromine

RESULTS

Oral Equivalent Dose Table

Oral Equivalent Dose Plot

Bioactivity Exposure Ratio Table

Bioactivity Exposure Ratio Plot

Download Figure 2

Bioactivity Exposure Ratio (BER) for Simulated Chemicals

Compound	Bioactivity Exposure Ratio (BER)
Saccharin	~10 ^{-3.5}
Benzophenone	~10 ^{-0.5}
Eugenol	~10 ²
Theobromine	~10 ⁴

Figure 2: Plot of the bioactivity exposure ratio (BER) values calculated from the oral equivalent dose and exposure estimate from each chemical. The red dotted line (BER = 1) separates compounds with BER < 1, where these chemicals are suggested for prioritization.

Try It On Your Own



Try your own simulation in the IVIVE module

OR

Obtain the human bioactivity exposure ratios for the following chemicals

Chemical Name	CASRN	Bioactive Concentration (uM)	Exposure (mg/kg-BW/day)
FD&C Red 40	25956-17-6	11.58	6.77
Carminic acid	1260-17-9	4.365	7.39E-3
FD&C Green 3	2353-45-9	5.998	1.40E-4
FD&C Yellow 6	2783-94-0	12.59	12.1

and use the following inputs:

- Use the pbtk model and the Honda1 IVIVE condition
- Use the 90th quantile steady state concentration (10th OED quantile)
- Allow the use of in silico parameters

Questions?